

THE ACTUATOR PETTING ZOO

Hands-on multimodal prototyping · Week 39 lab · 2 hours

WHY WE ARE HERE

You have had the actuator lecture. You know what an ERM is on a slide.

Today you put one against your own wrist, decide it feels cheap, and pick something else — before you commit a project to it.

Every year projects arrive at the mid-way pitch built entirely on the screen and the speaker.

Not because haptics was the wrong choice — because nobody knew a coin motor costs 12 kr and takes two wires.

HOW TODAY WORKS

Six benches. Already wired. Already running.

- Ten minutes each, on a hard timer.
- You change one parameter and answer one question.
- You wire nothing — that is what the project weeks are for.

Then you build one tacton and someone else tries to read it.

RUN SHEET

Times are offsets from the start. The rotation is the spine — keep it moving.

0:00	10 min	Framing, and a demo that fails
0:10	60 min	Bench rotation — 10 minutes each
1:10	10 min	Round-the-room: what surprised you
1:20	30 min	Build one tacton
1:50	10 min	Blind test and pack down

THE SIX BENCHES

01 — ERM COIN MOTOR

CUTANEOUS · VIBRATION

The eccentric rotating mass from Lecture 6. The same part that is in your phone and every game controller.

Turn the pot and notice what you cannot do: frequency and amplitude are mechanically welded together. "Gentle but fast" is not available to you.

part 10mm 3V coin ERM
drive 2N2222 + 1N4148
pin D9 (PWM)
draw ~75 mA
spin-up 20–40 ms
cost ~12 kr

CARD QUESTION

At what point does it stop feeling like information and start feeling like a malfunction?

02 — LRA + PIEZO DISC

CUTANEOUS · VIBRATION

Two ways out of the ERM's compromise.

The LRA hits resonance in ~5 ms and stops just as fast — which is why phone keyboards use it.

The piezo is the crystal from the lecture: near-instant, almost no power, but shallow. A tick, not a thump.

parts LRA 10mm + piezo
driver DRV2605L (I2C)
pin A4 SDA / A5 SCL
effects 123 waveforms
cost ~90 kr

CARD QUESTION

Which of the three vibrators would you actually put on a wrist, and why not the others?

03 — SOLENOID TAP

CUTANEOUS · IMPACT

A push-pull solenoid firing a single 15 ms pulse against a fingertip.

This is the bench everyone remembers. A discrete knock carries urgency no amount of buzzing does — it reads as somebody tapping you.

Indexical, not symbolic.

part	5V push-pull
drive	MOSFET + flyback
pin	D6
peak	~1.1 A
duty	<= 25%, gets hot
cost	~45 kr

CARD QUESTION

How many taps before it goes from alert to nagging?

04 — CAPACITIVE TOUCH

KINESTHETIC · TOUCH-TO-ACTUATE

Touch a surface with a bare finger and the servos react.

There is no button and no sensor you can point at. The input IS a wire taped to a plate, read on an analogue pin.

This is the SOS 2025 rig running its real sketch.

```
sense    ADCTouch on A0
baseline  25-sample roll
trigger   1.01x average
debounce  100 ms
servos    D4 + D13
cost      ~0 kr – a wire
```

CARD QUESTION

Put your hand near the plate without touching. When exactly did it decide that was a touch?

BENCH 04 — THE WHOLE TRICK

Capacitive sensing costs a wire. What it does not cost you is thinking.

reading > baselineAverage() * 1.01

The threshold is RELATIVE, never absolute.

- Readings drift with humidity, mains hum, how you are sitting. A fixed threshold works in the morning and fails after lunch.

The baseline only learns while untouched.

- Otherwise a long press slowly teaches it that a finger is normal, and the touch silently stops registering.

05 — PELTIER WARM / COOL

CUTANEOUS · THERMAL

A thermoelectric tile: heats one side, cools the other, reverses when you flip the current.

Sit with it. It takes several seconds before you are sure which way it went.

The lecture's warning holds up in person — alternating hot and cold just reads as lukewarm.

part TEC1-12706
drive L298N H-bridge
pin D3 PWM / D4-5
draw 2–4 A, bench PSU
onset 3–8 s
cap 45 C in software

CARD QUESTION

Time yourself. How long until you would bet money on warmer vs cooler?

06 — THE SAME SIGNAL, TWICE

INTRAMODAL · AUDIO + HAPTIC

One transducer on a plate, playing 40 Hz you feel and 400 Hz you hear
— from the same driver.

Toggle between them alone and together.

Together is not louder. It is more CERTAIN. That is redundancy from
Lecture 7, and the cheapest reliability win your project can get.

part	bone-conduction
amp	PAM8403 class-D
pin	D11 (tone)
felt	~40 Hz
heard	~400 Hz
cost	~110 kr

CARD QUESTION

With ear defenders on, does the tacton still work?

INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table, not a rotation. Wander over during the build.

- Capacitive touch — a wire on an analogue pin. See bench 04.
- Heart rate (PPG) — steady at rest, garbage the moment the hand moves.
- GSR — arousal, but it drifts. Relative change only, never absolute.
- Flex sensor — the cheap route to a data glove.
- Pressure (FSR) — how hard, not just whether. Calibrate per pad.
- Your own phone — accelerometer, gyro, mic, camera, vibration. No soldering at all.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Capacitive sensing + servos + actuators — SOS 2025, Arduino Uno

- A moving screen that breathes when idle and retreats when touched. 922 lines, plus a browser control panel over serial.
- Take away: easing and idle motion are what separate "a servo moved" from "it reacted to me". Both are a dozen lines.

Instrumented sock — bachelor project, ESP32

- Six force sensors under a foot, batched to a server over Wi-Fi, each with its own calibration constants.
- Take away: calibrate per sensor, and batch your writes.

BUILD ONE TACTON

THE BRIEF

30 minutes. Deliberately not long enough to be precious about it.

Pick one actuator from the rotation.

Encode three messages: arrived / something wrong / finished.

Vary RHYTHM and ROUGHNESS only — the Lecture 7 tacton parameters. No swapping actuators between messages.

Hand it to another group with the display turned away. They name all three.

Write down which two got confused, and what would have separated them.

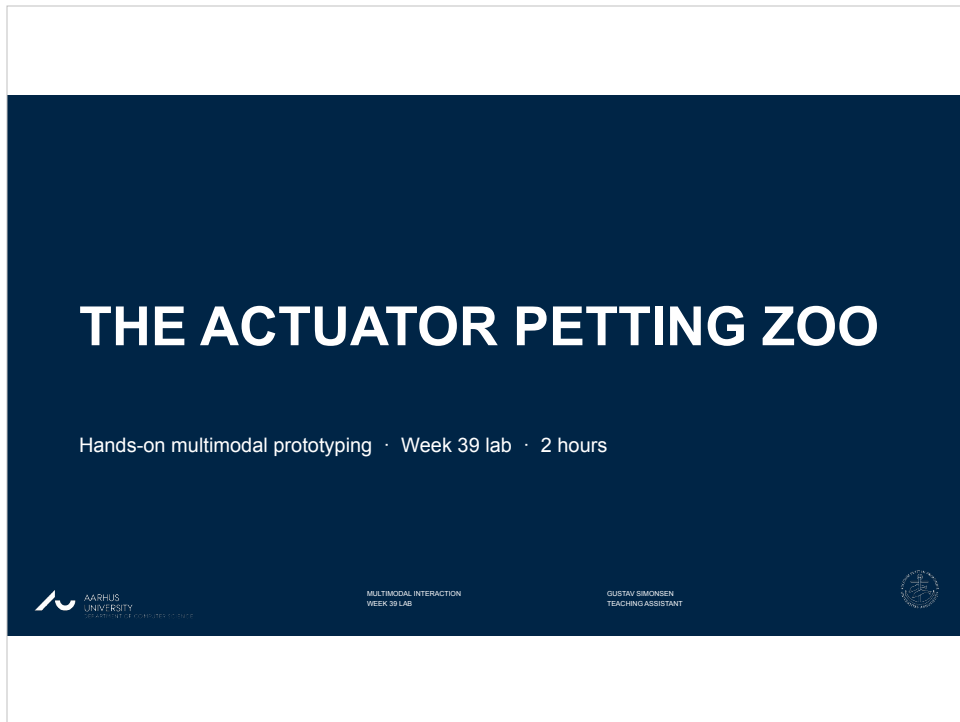
THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If your design only works when someone is watching a screen, you built a visual interface with a motor glued to it.

GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop



0:00 — Start here. Don't read the slide.

Open with the failing demo, not with talking.
Have the coin motor and the solenoid both wired and ready before anyone sits down.

Buzz a phone-style notification on the ERM.
Then send the same message on the solenoid. Ask: which felt more urgent? Let them answer. Nobody needs theory for that, and that IS the argument for the whole session.

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Every year projects arrive at the mid-way pitch built entirely on the screen and the speaker.

Not because haptics was the wrong choice — because nobody knew a coin motor costs 12 kr and takes two wires.



Say the honest version: every year projects end up on the screen and the speaker, and haptics gets designed on paper then dropped.

Not because it was wrong. Because nobody had touched the parts.

Keep this short — 2 minutes. They came to do things, not hear framing.

HOW TODAY WORKS

Six benches. Already wired. Already running.

- Ten minutes each, on a hard timer.
- You change one parameter and answer one question.
- You wire nothing — that is what the project weeks are for.

Then you build one tacton and someone else tries to read it.



Set the rules firmly, because the rotation collapses if you don't.

Nobody wires anything today. Everything is pre-wired and running. They turn one knob and answer one question.

Say out loud: the timer is hard. Ten minutes means ten minutes. Bench 3 (the solenoid) is the one people won't want to leave.

RUN SHEET

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Show it, don't narrate it. Fifteen seconds.

Practical: assign each group a DIFFERENT starting bench so they rotate cleanly. Six groups, six benches. Write starting numbers on the whiteboard.

If a bench breaks mid-session, pull it and go to five. Five working benches beat six with a mystery.



0:10 — Rotation starts. Set a visible timer.

You are now a timekeeper, not a lecturer. Call the rotation loudly every ten minutes. Circulate, but don't rescue — let them poke at things.

The bench slides that follow are reference for YOU and for anyone who wants to look back. Do not present them one by one before the rotation. That would eat the hour.

01 — ERM COIN MOTOR

CUTANEOUS · VIBRATION

The eccentric rotating mass from Lecture 6. The same part that is in your phone and every game controller.

Turn the pot and notice what you cannot do: frequency and amplitude are mechanically welded together. "Gentle but fast" is not available to you.

part 10mm 3V coin ERM
drive 2N2222 + 1N4148
pin D9 (PWM)
draw ~75 mA
spin-up 20–40 ms
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CARD QUESTION

At what point does it stop feeling like information and start feeling like a malfunction?



ERM. The lecture point made physical: frequency and amplitude are mechanically linked. You cannot get gentle-but-fast.

Common question: "why does it feel weak at low PWM?" Below about 90 the motor can't overcome its own friction. That floor is real and worth naming.

Never let anyone rewire this to a bare pin — 40 mA limit, motor pulls 75.

02 — LRA + PIEZO DISC

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Two ways out of the ERM's compromise.

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CARD QUESTION

Which of the three vibrators would you actually put on a wrist, and why not the others?



The escape from bench 1's compromise. LRA is the phone-keyboard feel: 5 ms on, 5 ms off. Piezo is sharp but shallow.

If the LRA sounds wrong, the DRV2605L is in ERM mode — it must be told which type it's driving. That's the failure to expect here.

Good moment to point out: 123 built-in waveforms means they don't have to design vibration from scratch in their project.

03 — SOLENOID TAP

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This is the bench everyone remembers. A discrete knock carries urgency
no amount of buzzing does — it reads as somebody tapping you.

Indexical, not symbolic.

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pin D6
peak ~1.1 A
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CARD QUESTION

How many taps before it goes from alert to nagging?



The memorable one. A knock reads as a person tapping you — indexical, not symbolic, straight from Lecture 7's semiotics.

Watch the duty cycle. Solenoids get hot fast; if it's been hammering for ten minutes, let it rest.

Separate supply, common ground. On USB alone the board browns out and resets, which looks like a code bug and isn't.

04 — CAPACITIVE TOUCH

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CARD QUESTION

Put your hand near the plate without touching. When exactly did it decide that was a touch?



The surprise bench. No breakout board — the input is a wire.

Have the Serial Plotter on a screen here.
Watching the baseline wander while
the trigger line tracks it teaches the whole idea
in ten seconds.

Let someone hold their hand NEAR without
touching. The gradual approach is
what makes the threshold question real.

BENCH 04 — THE WHOLE TRICK

Capacitive sensing costs a wire. What it does not cost you is thinking.

`reading > baselineAverage() * 1.01`

The threshold is RELATIVE, never absolute.

- Readings drift with humidity, mains hum, how you are sitting. A fixed threshold works in the morning and fails after lunch.

The baseline only learns while untouched.

- Otherwise a long press slowly teaches it that a finger is normal, and the touch silently stops registering.



Slow down here. This is the one idea most worth stealing from today.

Two things students get wrong:

1. They use a fixed threshold. Works in the morning, fails after lunch,
because the reading drifts with humidity and mains hum.

2. They feed the baseline buffer always. Then a long press teaches the
baseline that a finger is normal, and the touch silently stops working.

The buffer must ONLY learn while untouched

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CARD QUESTION

Time yourself. How long until you would bet money on warmer vs cooler?

 AARHUS
UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
WEEK 39 LAB

GUSTAV SIMONSEN
TEACHING ASSISTANT



Thermal. The lecture said people are bad at thermal patterns — this is where they feel why.

Make them time themselves. The 3-8 second onset kills a lot of bad project ideas before they get proposed, which is exactly what you want happening now rather than in week 47.

Safety: heatsink on the hot side, capped at 45 C. Skin burns above 50.

06 — THE SAME SIGNAL, TWICE

INTRAMODAL - AUDIO + HAPTIC

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— from the same driver.

Toggle between them alone and together.

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Lecture 7, and the cheapest reliability win your project can get.

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CARD QUESTION

With ear defenders on, does the tacton still work?



Redundancy, made physical. Same driver, two frequencies — one you feel, one you hear.

The point isn't that together is louder. It's that together is more CERTAIN.

That's crossmodal redundancy from Lecture 7 and it's the cheapest reliability win available to their projects.

Ear defenders must actually be used or the comparison is dishonest.

INPUTS, AND PRIOR ART



1:10 — Rotation over. Round-the-room now.

One sentence per group: what surprised you.
No slides, no laptops. Ten minutes.

Expect the thermal group to report the delay.
Let that land — it's more
persuasive coming from a peer than from you.

INPUTS WORTH KNOWING ABOUT

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- GSR — arousal, but it drifts. Relative change only, never absolute.
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- Pressure (FSR) — how hard, not just whether. Calibrate per pad.
- Your own phone — accelerometer, gyro, mic, camera, vibration. No soldering at all.



Mention, don't teach. These sit on one table with a sign.

The honest warnings matter more than the specs: PPG is garbage once the hand moves, GSR drifts so only relative change is usable, FSRs need per-pad calibration.

Push the phone option hard. For a lot of projects it genuinely beats everything else on the table and needs no soldering.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

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- A moving screen that breathes when idle and retreats when touched. 922 lines, plus a browser control panel over serial.
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Instrumented sock — bachelor project, ESP32

- Six force sensors under a foot, batched to a server over Wi-Fi, each with its own calibration constants.
- Take away: calibrate per sensor, and batch your writes.



Five minutes, and show the actual repos on screen if you can.

The argument: this was built by students at your stage, not by a lab with a budget. The gap between "we should use haptics" and "it moves when you touch it" is smaller than it looks.

Note for you: the sock repo has a hardcoded Wi-Fi password and live patient IDs in it. Scrub before showing, or show only the calibration section.



BUILD ONE TACTON



1:20 — Build starts. Thirty minutes.

Groups pick any actuator from the rotation.
Most will pick bench 1 or 3
because they're simplest — that's fine, the
constraint does the work.

THE BRIEF

30 minutes. Deliberately not long enough to be precious about it.

Pick one actuator from the rotation.

Encode three messages: arrived / something wrong / finished.

Vary RHYTHM and ROUGHNESS only — the Lecture 7 tacton parameters. No swapping actuators between messages.

Hand it to another group with the display turned away. They name all three.

Write down which two got confused, and what would have separated them.



Read the brief out once, then get out of the way.

Rhythm and roughness only. If someone asks to add a second actuator, say no — the constraint IS the exercise. Encoding within one channel is the skill.

Suggested set is on the slide, but let them choose their own three messages if they have a better idea.

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If your design only works when someone is watching a screen, you built a visual interface with a motor glued to it.



This is the line worth repeating twice.

1:50 — Blind test. Swap groups, receiver looks away, guess all three.

Then have each group name which two got confused and what would have separated them. That confusion is the actual learning, not the successful ones.

Pack down: everything back in the box, including the resistors. It will not all come back.



Point them at the repo. Bench 01 and 04 sketches are commented for them.

Last thing to say: pick your actuator before you design the interaction, not after. Today was so they can do that from experience rather than a datasheet.

If anyone wants to borrow parts for the project weeks, now is when they ask.